

Jordan Form 1

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Problem 1. We say that non-zero v is a "generalized eigenvector" for A if $(A - fI)^k v = 0$ for some f in F and some $k > 0$. Show that f is unique, ie if v in the kernel of $(A - fI)^s$ and $(A - gI)^t$, then $f = g$. Hint: Let $S = A - fI, T = A - gI$ explain why some high power of $S - T$ will kill v . Conclude $f = g$.

Proof. Suppose $v \in \ker((A - fI)^s) \cap \ker((A - gI)^t)$. Let $S = A - fI$ and $T = A - gI$. Let $r = s + t$. $ST = (A - fI)(A - gI) = A^2 - gAI - fIA + fgI^2 = A^2 - (g + f)A + fgI$ and $TS = (A - gI)(A - fI) = A^2 - fIA - gAI + fgI^2 = A^2 - (g + f)A + fgI$. Thus $ST = TS$. Since S and T commute we can use the binomial theorem. Thus $(S - T)^r = \sum_{n=0}^r \binom{r}{n} S^n T^{r-n}$. For all n such that $r - n \geq t$, $T^{r-n} v = T^{r-n-t} T^t v = T^{r-n-t}(0) = 0$. For all n such that $r - n < t$, then $S^n T^{r-n} = T^{r-n} S^n$ (by commutativity above) and $s + t - n < t$ and $s - n < 0$ thus $n > s$ thus $S^n v = S^{n-s} S^s v = S^{n-s}(0) = 0$ thus $T^{r-n} S^n v = 0$. Thus $\forall n \in [0, \dots, r]$ $S^n T^{r-n} v = 0$. Thus $(S - T)^r v = \sum_{n=0}^r \binom{r}{n} S^n T^{r-n} v = \sum_{n=0}^r \binom{r}{n} S^n T^{r-n} v = 0$. However $S - T = (g - f)I$. Then $((g - f)I)^r v = (g - f)^r I v = (g - f)^r v = 0 = 0v$. Thus $(g - f)^r = 0$ and $g - f = 0$ and $g = f$. $\square$

Problem 2a. In fact, we can take $k = \dim(V)$: Let $S = A - fI$, and $W_i := \ker S^i$. Explain why these subspaces "increase" with i , ie we have $0 \subset W_1 \subset W_2 \subset \dots$

Proof. Base Case: Let $k = 1$. Then by definition of a subspace, $0 \subset W_1$.

Inductive Hypothesis: Assume that $W_k \subset W_{k+1}$

Inductive Step: Let $W_{k+2} := \ker S^{k+2}$. $\forall v \in W_{k+1}$ then $S(S^{k+1}v) = S^{k+2}v$ but since $v \in \ker S^{k+1}$ then $S^{k+1}v = 0$ thus $S^{k+2}v = 0$. Thus $v \in W_{k+2}$. Thus $W_{k+1} \subset W_{k+2}$.

Thus by induction, $0 \subset W_1 \subset W_2 \subset \dots$ $\square$

Problem 2b. Show that if $W_i = W_{i+1}$ then $W_{i+1} = W_{i+2}$, and thus all the W_k for $k \geq i$ are the same.

Proof. Suppose $W_i = W_{i+1}$. Base Case: By the previous proof we know $W_{i+1} \subset W_{i+2}$. $\forall v \in W_{i+2}$, Thus $S^{i+2}v = S^{i+1}(Sv)$, thus $Sv \in \ker S^{i+1} = W_{i+1} = W_i = \ker S^i$. Thus $Sv \in \ker S^i$. Thus $S^i Sv = S^{i+1}v = 0$ and $v \in W_{i+1}$. Thus $\forall v \in W_{i+2}$ then $v \in W_{i+1}$. Thus $W_{i+1} \subset W_{i+2}$ and $W_{i+1} = W_{i+2}$.

Inductive Hypothesis: For a positive integer $k > 0$, assume $W_{i+k} = W_{i+k+1}$.

Inductive Step: By the previous proof we know $W_{i+k+1} \subset W_{i+k+2}$. $\forall v \in W_{i+k+2}$. By definition of kernel, $S^{i+k+2}v = 0 = S^{i+k+1}Sv = 0$. Thus $Sv \in W_{i+k+1}$. By the inductive hypothesis, since $W_{i+k+1} = W_{i+k}$. Thus $Sv \in W_{i+k}$ thus $S^{i+k}Sv = S^{i+k+1}v = 0$. Thus $v \in W_{i+k+1} = W_{i+k}$. Thus $W_{i+k+2} = W_{i+k+1} = W_{i+k}$.

Thus by induction, $\forall k \in \mathbb{Z}$ s.t $k > 0$ then $W_i = W_{i+k}$. $\square$

Problem 2c. Now explain why $W_k = W_{\dim V}$ for sufficiently big k , and conclude the generalized eigenvectors are the (non-zero) vectors in $\ker(A - fI)^{\dim V}$. Now we define the generalized eigenspace, for f , to be $GV_f(A) := \ker(A - fI)^{\dim V}$.

Proof. Since $W_i \subseteq V$ are subspaces of V , $\forall i \in \mathbb{N}$. The $\dim W_i \leq \dim V$. Thus $\dim W_{\dim V} \leq \dim V$. If $\dim W_{\dim V} < \dim V$, then there is a $0 \leq k < \dim V$ s.t $W_k = W_{k+1}$ because otherwise $\dim W_{k+1} > \dim W_k$ for all k . and then $\dim W_{\dim V} = \dim V$ forming a contradiction with the assumption. Thus $\exists k$ s.t $W_k = W_{k+1}$. By previous proof, $W_k = W_{\dim V}$ because $\dim V > k$. However if $\dim W_{\dim V} = \dim V$ and since $W_{\dim V} \subseteq W_{\dim V+1}$ thus $\dim W_{\dim V} \leq \dim W_{\dim V+1} \leq \dim V$ thus $\dim W_{\dim V+1} = \dim V$ thus $\dim W_{\dim V} = \dim W_{\dim V+1}$ and $W_{\dim V} = W_{\dim V+1}$. Thus $k = \dim V + 1$. By the previous theorem, $\forall n \geq k$ $W_{\dim V} = W_n$.

Thus $W_{\dim V}$ contains all W_i for $i \leq \dim W$ and is equal to all W_i for $i \geq \dim W$. Thus all the generalized eigenvectors are the nonzero vectors in $W_{\dim V}$. $\square$

Problem 2d. Show this is non-zero iff f is an eigenvalue.

Proof. ($\implies$): $GV_f(A) \neq 0$.

1. If there exists no $k \in [1, \dots, \dim V]$ such that $\exists v \in GV_f(A)$ s.t $v \notin W_k$ then $W_1 = GV_f(A)$. Thus since $GV_f(A) \neq 0$, then $W_1 \neq 0$ W_1 is by definition a eigenspace of A . Thus f must be a eigenvalue.
2. If there exists a $k \in [1, \dots, \dim V]$ such that $\exists v \in GV_f(A)$ s.t $v \notin W_k$. Thus $\forall v \in W_k$, $v \notin W_{k-1}$ because if otherwise $k = k-1$ which is a contradiction. Thus $\forall v \in W_k$, $(A - fI)^k v = 0$ and $(A - fI)((A - fI)^{k-1}v) = 0$. Since $(A - fI)^{k-1}v \neq 0$, then f must be a eigenvalue by definition.

($\Leftarrow$): Suppose f is a eigenvalue. Thus there is a nonzero eigenvector v such that $(A - fI)v = 0$ by definition of eigenvalues and eigenvectors. Thus $v \in W_1$. Since $W_1 \subset W_{\dim V}$ by 2a. $v \in W_{\dim V} = GV_f(A)$. Thus $GV_f(A) \neq 0$ $\square$

Problem 3. What do you expect is true about the generalized Eigenspaces? Lets say $GV_f(A)$ is the kernel of $(A - fI)^{\dim V}$. What do you think is true of the map $\oplus_{f \in F} GV_f(A) \rightarrow V$ (the addition map). Prove it – the argument is similar to the eigenspace case, just a bit more involved.

Solution. Since $GV_f(A) \subset V$ is a subspace and thus $\dim(GV_f(A)) \leq \dim V$. $\blacksquare$

Claim 3. $\oplus_{f \in F} GV_f(A) \rightarrow V$ is injective.

Proof. Base Case (k = 2): For $f, g \in F$ where $f \neq g$, $\forall v \in GV_f(A)$ and $\forall w \in GV_g(A)$. Let $v + w = 0$. $A^{\dim(V)}(v + w) = 0$ and $A^{\dim(V)}v + A^{\dim(V)}w = 0$. Since $v \in GL_f(A)$ and $w \in GL_g(A)$. Thus $fv + gw = 0$. $f(v + w) = 0$. Thus $fv + fw = 0$. Thus $fv + gw - fv - fw = (g - f)w = 0$. But since $f \neq g$, then $g - f \neq 0$ thus $w = 0$. But since $v + w = 0$, then $v + 0 = 0$ and $v = 0$.

Inductive Hypothesis: If $\sum_{i=1}^k v_i = 0$ for $v_i \in GV_{f_i}(A)$ with f_i distinct implies $v_i = 0$.

Inductive Step: Suppose $\sum_{i=1}^{k+1} v_i = 0$ for $v_i \in GV_{f_i}(A)$ with f_i distinct. Then $\sum_{i=1}^{k+1} v_i = \sum_{i=1}^k v_i + v_{k+1} = 0$. By our inductive hypothesis, $0 + v_{k+1} = v_{k+1} = 0$.

Thus by induction $\sum_{i=1}^{k+1} v_i = 0 \implies v_i = 0$. Thus the kernel of $\oplus_{f \in F}$ is only the 0 vector. Thus the $\oplus_{f \in F}$ is injective. Thus $\oplus_{f \in F} GV_f(A) \subset V$ and $\dim(\oplus_{f \in F} GV_f(A)) \leq \dim V$ $\square$

Problem 4. Now let's prove that this map is always an isomorphism, ie V is always the direct sum of the generalized eigenspaces (assuming that the characteristic polynomial has all its roots in F). To do this use the Cayley-Hamilton theorem (which we will prove next term, if not before then): namely that any matrix satisfies its characteristic polynomial, together with TKL (that kids lemma, about the dim of the kernel of a product).

Proof. The generalized eigenvalues of A is $f_1, \dots, f_n$.

The characteristic polynomial where $m = \dim V$, $p_A(x) = (x - f_1)^m \dots (x - f_n)^m$. By Cayley Hamilton Theorem, $p_A(A) = 0$. Thus $\ker p_A(A) = V$ thus $\dim \ker p_A(A) = \dim V$. By TKL, $\dim V = \dim \ker p_A(A) \leq \dim \ker(x - f_1)^m + \dots + \dim \ker(x - f_n)^m = \dim GV_{f_1}(A) + \dots + \dim GV_{f_n}(A)$. Thus $\dim V \leq \dim GV_{f_1}(A) + \dots + \dim GV_{f_n}(A)$. But $\oplus_{f \in F} GV_f(A)$ is a subspace of V , thus $\dim V \geq \dim GV_{f_1}(A) + \dots + \dim GV_{f_n}(A)$. Thus $\dim V = \dim GV_{f_1}(A) + \dots + \dim GV_{f_n}(A)$. Thus $\oplus$ is a bijection and thus an isomorphism.

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